

Prakash Madai

Concord, NH | prakash.madai17@gmail.com | linkedin.com/in/prakashmadai10 | [Portfolio](#)

Education

Master of Science, Computer Science

University of Texas Permian Basin (UTPB), Odessa, TX | GPA: 3.9/4

Aug 2022 – May 2024

Bachelor's, Computer Engineering

Tribhuvan University, Kathmandu, Nepal

Dec 2014 – Dec 2018

Skills

- **ArcGIS Ecosystem:** ArcGIS Enterprise (Server, Portal, Data Store, Web Adaptor, SDE), ArcGIS Pro, CityWorks, ArcGIS Online, Hub, Solutions, Dashboards, Experience Builder, StoryMaps
- **Automation & Field Ops:** ArcPy, ArcGIS API for Python, Arcade, ModelBuilder, ETL Pipelines (Spatial/Non-Spatial), Geodatabase Design, Spatial Data Analysis, Field Maps, Survey123, Workforce, GPS, Georeferencing, Geocoding
- **Programming & Databases:** Python, R, MS SQL Server, Oracle, PostgreSQL, NoSQL (MongoDB), Database Design, Data Modeling, API, Linux, HTML, CSS
- **Data Engineering & AI:** Flask, Apache Spark, Airflow, SSIS, SSRS, MS Fabric, Power BI, Power Automate, Pandas, Matplotlib, Seaborn, AWS, AI/ML, Data Governance, QA/QC, Financial Services & Tech

Work Experience

GIS Solutions Engineer

State of New Hampshire, NH

04/2026 – Present

- Developed and maintained GIS web applications for fire-location triangulation and NH E9-1-1 address-management workflows using Python, Flask, FastAPI, ArcGIS JavaScript API, SQLAlchemy, and SQL Server
- Automated statewide GIS data processing in ArcGIS Pro using ArcPy, including addresses, roads, building footprints, emergency service zones, water layers, trails, AED locations, and address-locator creation
- Built automated ArcGIS Enterprise publishing workflows for tile packages and hosted basemap layers, including metadata updates, web map layer replacement, item-state tracking, logging, and failed-run recovery

GIS Data Analyst/Administrator

City of Midland, Texas

07/2025 – 01/2026

- Administered full ArcGIS Enterprise stack (Portal, Server, Data Store, SDE) and configured Hub collaboration for automated sync of 100+ layers
- Developed Python automation for metadata, permissions, and audit logs for 500+ items, cutting manual effort by ~90%
- Built SSIS ETL and ModelBuilder pipelines to automate nightly GIS refreshes and publish dashboards for City Council and zoning/permit departments
- Integrated EnerGov and Code Enforcement data into ArcGIS Dashboards to resolve geocoding inconsistencies and improve service responsiveness
- Delivered KPI datasets from public safety systems (RMS/CAD) to support police performance reporting and Fire Department field inspections

GIS Analyst/Administrator

City of Odessa, Texas

03/2024 – 06/2025

- Administered AGOL, ArcGIS Server, and SQL Server for 300+ users, managing 1,000+ layers and automating SharePoint backups via Python, SQL, and Arcade
- Led spatial analysis for the LCRR, analyzing 48,000+ utility records to produce compliance web maps, and deployed public apps like Fire Hydrant Locator
- Implemented ArcGIS Solutions for waste and signal management, enabling mobile data collection, field inspections, and real-time dashboard reporting

Data Analyst – Internship

Rocket Mortgage, Detroit, MI

05/2023 – 08/2023

- Developed AWS Glue ETL scripts using PySpark and implemented Apache HUDI for incremental data processing across S3 and SQL Server, optimizing storage costs and loan prioritization efficiency

Data Analyst & Technical Support

Nepal Clearing House Ltd, Nepal

01/2020 – 07/2022

- Engineered 35+ interactive dashboards and executive reports via ETL pipelines and OLTP/OLAP systems, analyzing over 100,000 records weekly to uncover critical financial trends
- Automated 1,000+ monthly CSV deliveries to financial institutions by integrating osTicket with ECC archives, reducing manual workflow effort by 80%
- Managed daily payment system operations across connectIPS and CORPORATEPAY platforms, ensuring maximum uptime and transaction reliability for national clearing services

Certifications & Training

Industry Certifications: ArcGIS Pro Foundation 2025 (Esri), CompTIA Data+ v2, Microsoft Fabric Analytics Engineer Associate, Lean Six Sigma Green Belt

Esri Instructor-Led Courses: ArcGIS Enterprise (Base Deployment, Administration Workflows, Python Scripting); Configuring Utility Networks